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NPTEL ONLINE CERTIFICATION COURSE

SUSTAINABLE TRANSPORTATION SYSTEMS

Lecture 33: Life Cycle Assessment (LCA): An Introduction

By-
Prof. Bhola Ram Gurjar
Dept. of Civil Engineering



Contents

- Life cycle assessment
  - Definition
  - Objectives of LCA
  - System boundaries
  - Stages of LCA
  - Goal & scope of LCA
  - Life cycle inventory
  - Impact assessment
  - Interpretation
  - Conclusion

Life Cycle Assessment (LCA): Definition

LCA is a systematic analysis of environmental impact over the course of the entire life cycle of a product, material, process, or other measurable activity.

LCA models the environmental implications of the many interacting systems that make up industrial production.

Life Cycle

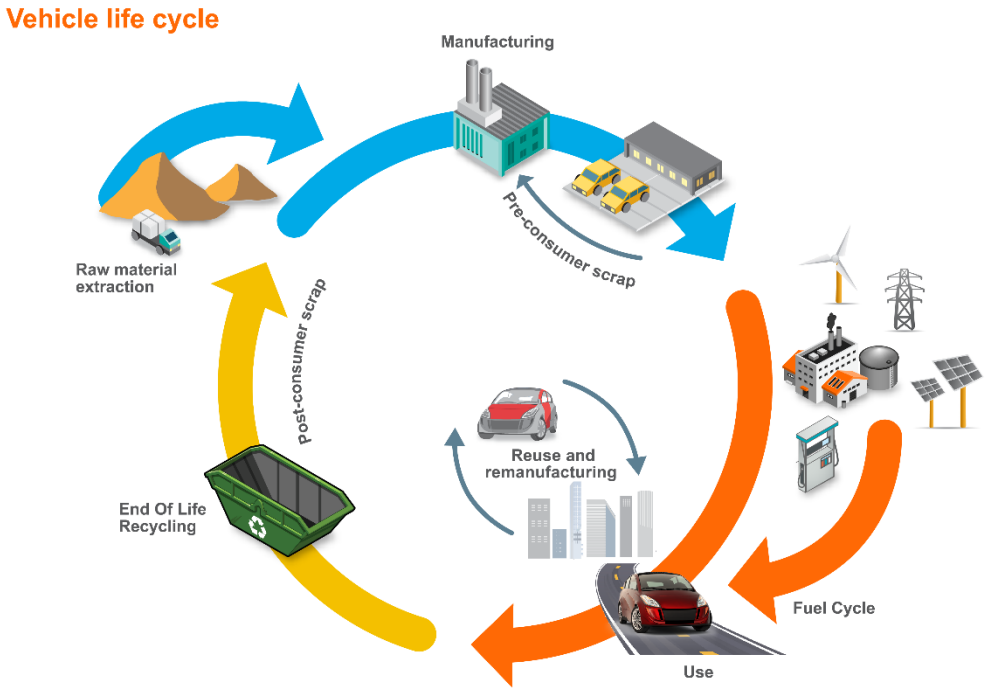


Image source: ahssinsights.org

Life cycle of a product or service is overall span of that product or service from extraction of raw material to application by user and further to waste and recycle.

Objectives of LCA: Product improvement

- Identifying opportunities to improve the environmental aspects of products/services at various points in their life cycle

Example:

If transport sector is major contributor in pollution in a city, we can assess the options whether we should use any alternative fuel or more polluting vehicle should be used for transport.

Objectives of LCA: Decision making

- Decision making in industry, governmental or non-governmental organizations (strategic planning, priority setting, product and process design or redesign)

Example:

Using BS-VI standard for vehicle In India from BS-IV standard vehicles to improve air quality in Indian cities.

Objectives of LCA: Environment planning

- Selection of relevant indicators of environmental performance, including measurement techniques

Example:

Deciding which parameters should be used for assessing the negative impacts on human health such as particulate matters, CO etc.

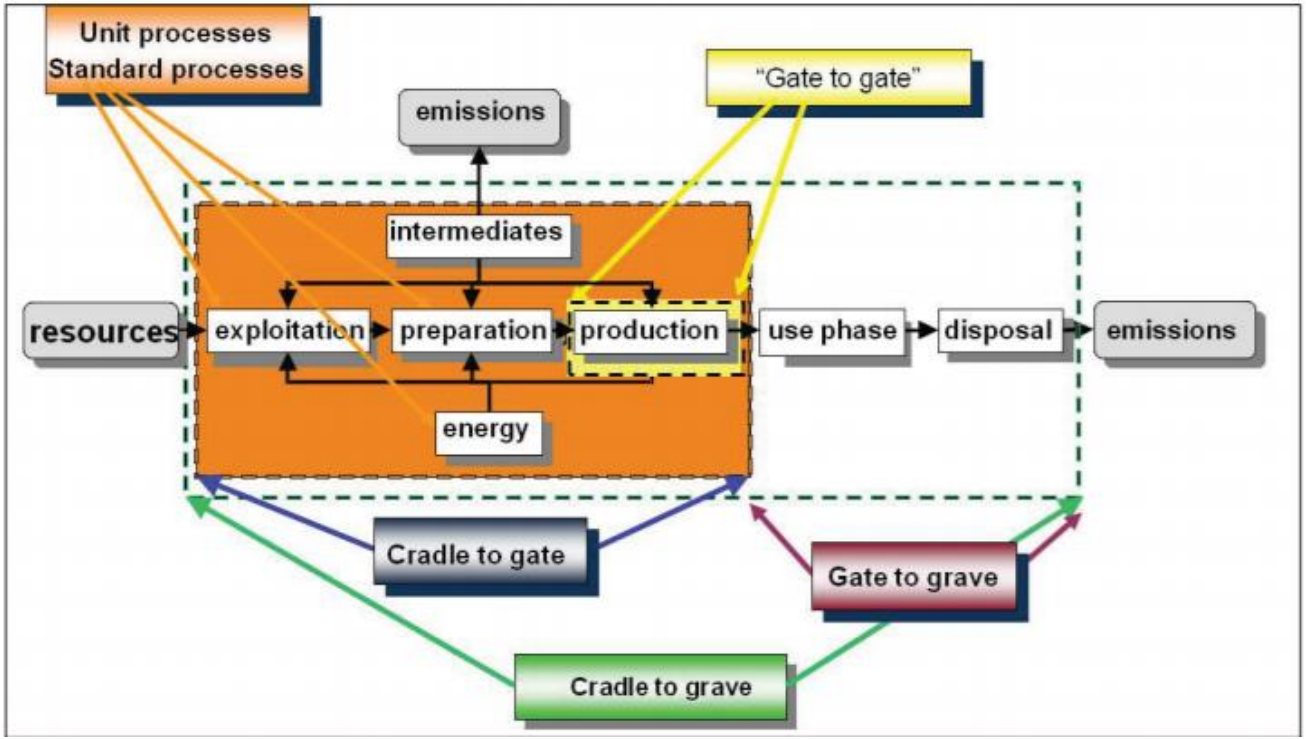
Objectives of LCA: Marketing

- Marketing (e.g., an environmental claim, eco-labeling scheme or environmental product declarations).

Example:

With widespread awareness about climate change, consumers prefer to use product with lest negative impacts on environment.
Companies may claim this for marketing backed up by LCA study.

System Boundaries



A system's boundaries are defined by cut-off criteria. Cut-off criteria are used to define the parts and materials included in and excluded from the product system.

Image : https://gabi.sphera.com/fileadmin/GaBi_Manual/GaBi_Paperclip_tutorial_Part1.pdf

Selection of System Boundaries

- Depending on the objective of study, system boundaries for LCA are defined. Often a combination of different cut-off criteria has to be used in order to define the system boundaries properly.
- These boundaries vary based on stages of life cycle considered for assessment
- There are four main options to define the system boundaries used:
  - Cradle to Grave
  - Cradle to Gate
  - Gate to Grave
  - Gate to Gate

Linear production



If a product manufacturing process starts from raw material extraction from natural resources to waste generation is considered in a linear fashion, it is called linear production model.

Image source: Galisano institute for sustainability
(https://www.rit.edu/sustainabilityinstitute/blog/what-life-cycle-assessment-lca)

System boundary: Cradle to grave

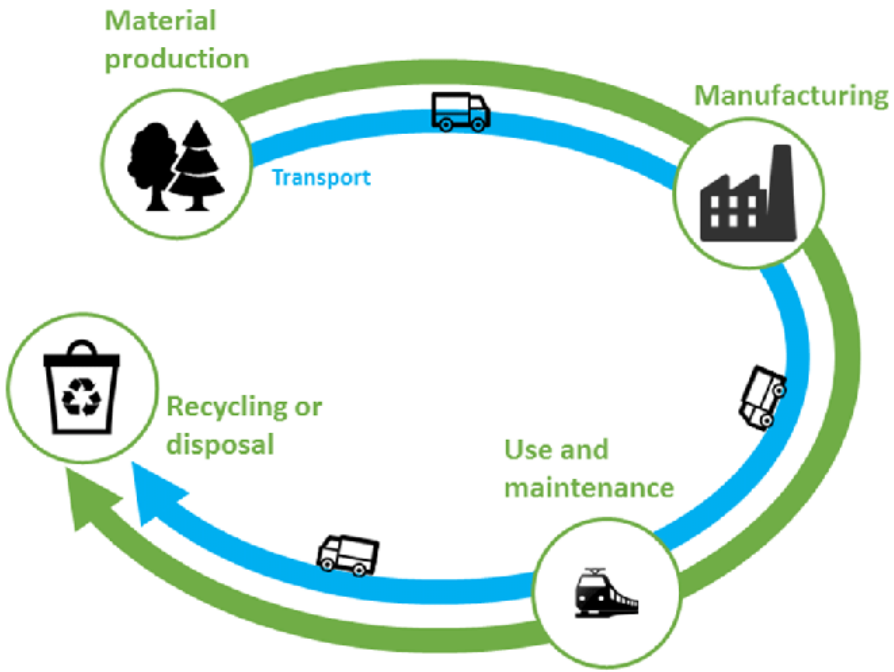


Image: Hegedic et al. 2018

If whole linear production model is considered for assessment during LCA from extraction of raw material, up to waste generation, it is called cradle to grave approach of LCA.

Circular production



- Production process including recovery of material after usage of product by user, is called circular production model.
- This is considered more environment friendly process compared to linear model.

Image source: Galisano institute for sustainability (https://www.rit.edu/sustainabilityinstitute/blog/what-life-cycle-assessment-lca)

System boundary: Cradle to cradle

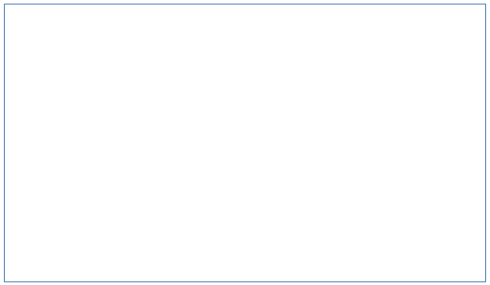
- Life cycle assessment done for whole circular production model is called cradle to cradle LCA approach.
- It is a variation of cradle-to-grave, exchanging the waste stage with a recycling process that makes it reusable for another/same product.
- It is also referred to as closed-loop LCA.



Image source: ecochain.com (https://ecochain.com/knowledge/life-cycle-assessment-lca-guide/)

System boundary : Cradle to gate

- It includes all processes from the raw material extraction through the production phase (gate of the factory)
- Cradle-to-gate only assesses a product until it leaves the factory gates, before it is transported to the consumer
- Used to determine the environmental impact of the production of a product



System boundary : Gate to grave & gate to gate

Gate to Grave: Assessment for the processes from the use and end-of-life phases (everything post-production)

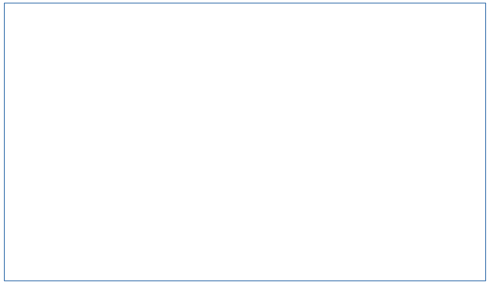
Used to determine the environmental impacts of a product once it leaves the factory.

Gate to Gate: Assessment for the processes from the production phase only

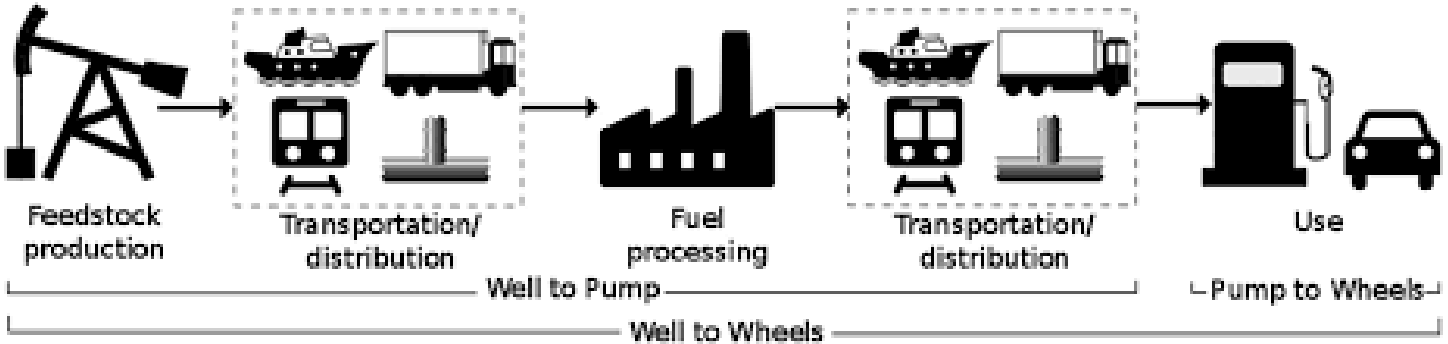
Used to determine the environmental impacts of production step or process.

There are two other concepts that are used for special requirements.

- Well-To-Wheel
- Economic Input-Output Life Cycle Assessment



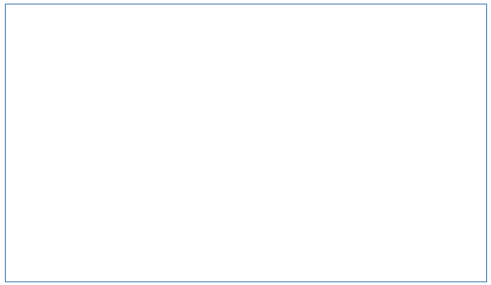
System boundary : Well to wheel



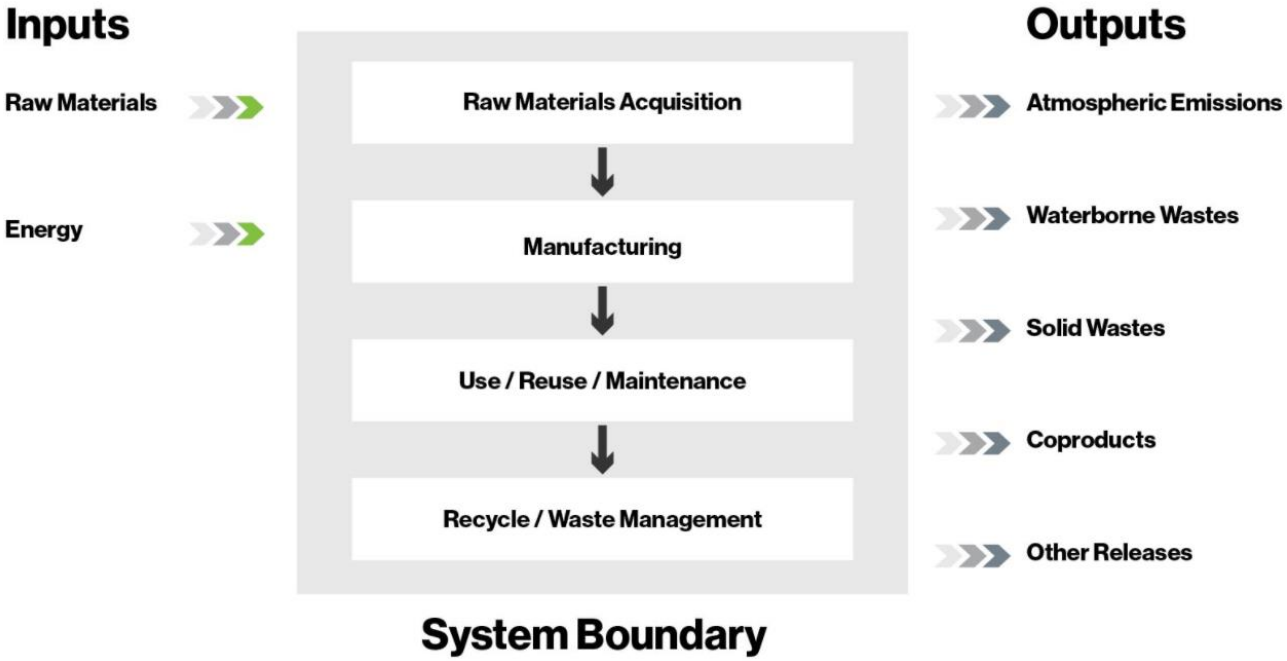
- *Well-to-wheel* is used for the Life Cycle Assessment of transport fuels and vehicles.
- This approach is more precise in calculating and assigning greenhouse gas emissions and energy usage for various fuels

System boundary : Economic Input-Output

- The EIOLCA (Economic Input-Output Life Cycle Assessment) aggregates industry data with the goal to create impact data for specific sectors.
- Averages are sometimes being used when no exact data is available.
- EIOLCA is not precise enough to make decisions on a product level.

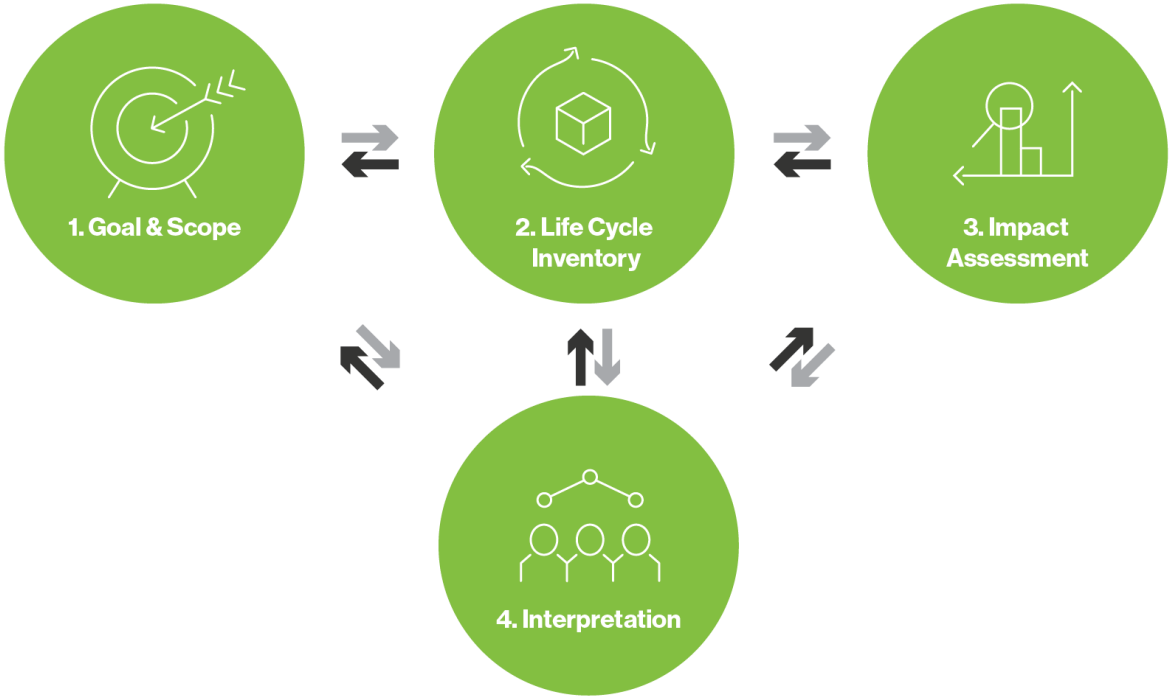


LCA process



Source: U.S. Environmental Protection Agency (1993)

Stages/Steps of LCA process



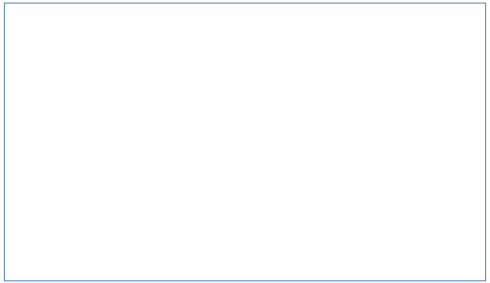
Stages/Steps of LCA:

- Defining goal & scope
- LC Inventory analysis
- Impact assessment
- Interpretation

Image source: Galisano institute for sustainability
(https://www.rit.edu/sustainabilityinstitute/blog/what-life-cycle-assessment-lca)

Scope of LCA

- Functional unit
- Reference flow
- Description of the system
- System boundaries
- Allocation procedures
- Impact categories and the impact assessment method
- Data requirements
- Data quality requirements



Functional unit & reference flow

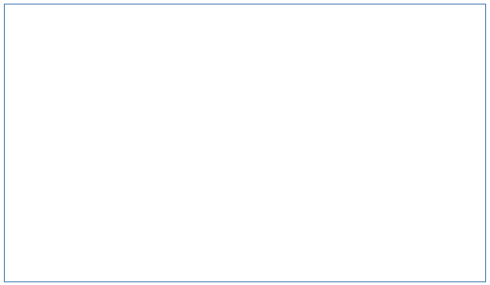
The functional unit is the quantified definition of the function of a product

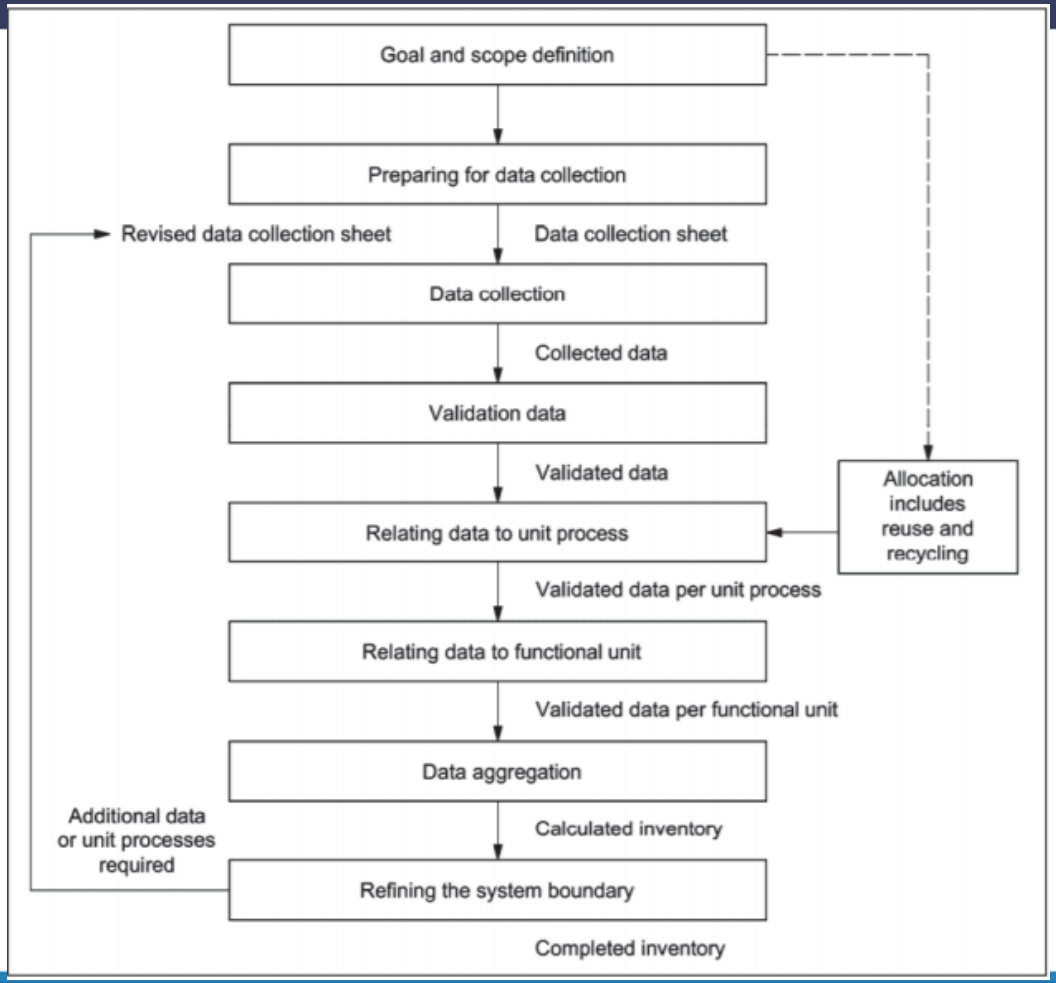
The reference flow is the measure of product components and materials needed to fulfill the function, as defined by the functional unit.

All data used in the LCA must be calculated or scaled in accordance with this reference flow.

Data requirement for LCA

- Energy inputs
- Raw material inputs
- Ancillary inputs
- Products & co-products
- Wastes
- Emissions to air, water and soil
- Other environmental aspects





Data collection process LC Inventory

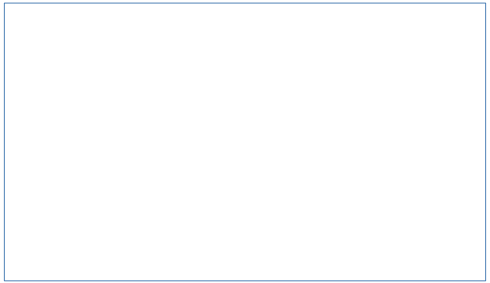
Source : https://gabi.sphera.com/fileadmin/GaBi_Manual/GaBi_Paperclip_tutorial_Part1.pdf

Factors affecting data quality 1/2

Data acquisition: Is the data measured, calculated or estimated? How much of the data required is primary data (in %) and how much data is taken from literature and databases (secondary data)?

Time-reference: When was this data obtained and have there been any major changes since the data collection that might affect the results?

Geographical reference: For what country or region is this data relevant?



Factors affecting data quality 2/2

- Technology: Is the secondary data from literature or databases representative for state-of-the-art-technology or for older technology?
- Precision: Is the data a precise representation of the system?
- Completeness: Are any data missing? How are data gaps filled?
- Presentative, consistency, reproducibility: Is the data representative, consistent and can it be reproduced?

Allocation classification

Allocation is the partitioning and relating of inputs and outputs of a process to the relevant products and byproducts.

Allocation by Mass: The inputs and outputs of a process are assigned to all its products proportionally to their mass

Allocation by Heating Value: The inputs and outputs of a process are assigned to all its products according to their heating value. This allocation method is often used for production processes of fuels.

Allocation by Market Value: The inputs and outputs of a process are assigned to all its products according to their market value

Life cycle inventory (LCI)

- An industrial system's inputs and outputs are measured and recorded (according to the functional unit).
- By the end of this phase, an inventory list is created that details all input/output data for the system under study.

Life cycle inventory: Flow model

- Collecting the input and output data within a list or table would quickly lead to confusion.
- That is why the Life Cycle Inventory is typically illustrated with a flow model.



Image source: ecochain.com (https://ecochain.com/knowledge/life-cycle-assessment-lca-guide/)

Life Cycle Inventory (LCI): Example of cement

Metric	Cradle-to-gate total per metric tonne of production	Unit
Environmental impact		
Global warming potential (100 years)	1040	kg CO2-eq.
Acidification potential	2.45	kg SO2-eq.
Eutrophication potential	1.22	kg N-eq.
Formation potential of tropospheric ozone	48.8	kg O3-eq
Ozone depletion potential	2.61E-05	kg CFC 11-eq.
Total primary energy consumption		
Non-renewable primary energy: Fossil	5250	MJ
Non-renewable primary energy: Nuclear	345	MJ
Renewable primary energy: Solar, wind, hydroelectric, geothermal	127	MJ
Renewable primary energy: Biomass	165	MJ

Impact Assessment

Evaluating the inventory of data that has been collected in order to make it meaningful in the context of potential damage to the environment or human health.

Various impacts are assessed in categorized manner (in various impact category).

Impact Assessment: Impact Category

- Global warming: Rise in global mean temperature due to Green House Gas(GHG) emissions
- Acidification: Due to excess CO2 emission, PH of ocean decreasing and affecting marine life
- Eutrophication: Excess algae in lake/water body which impacts the ecology
- Smog creation: Impacts human health
- Ozone depletion: Ozone layer in atmosphere depletes and filtration of Ultra Violate rays decreases causing skin diseases

Life Cycle Impact Assessment (LCIA): Steps

The U.S. Environmental Protection Agency breaks down an LCIA into the following steps:

- Selection and definition of impact categories
- Classification
- Characterization
- Normalization
- Grouping
- Weighting
- Evaluating and reporting LCIA results

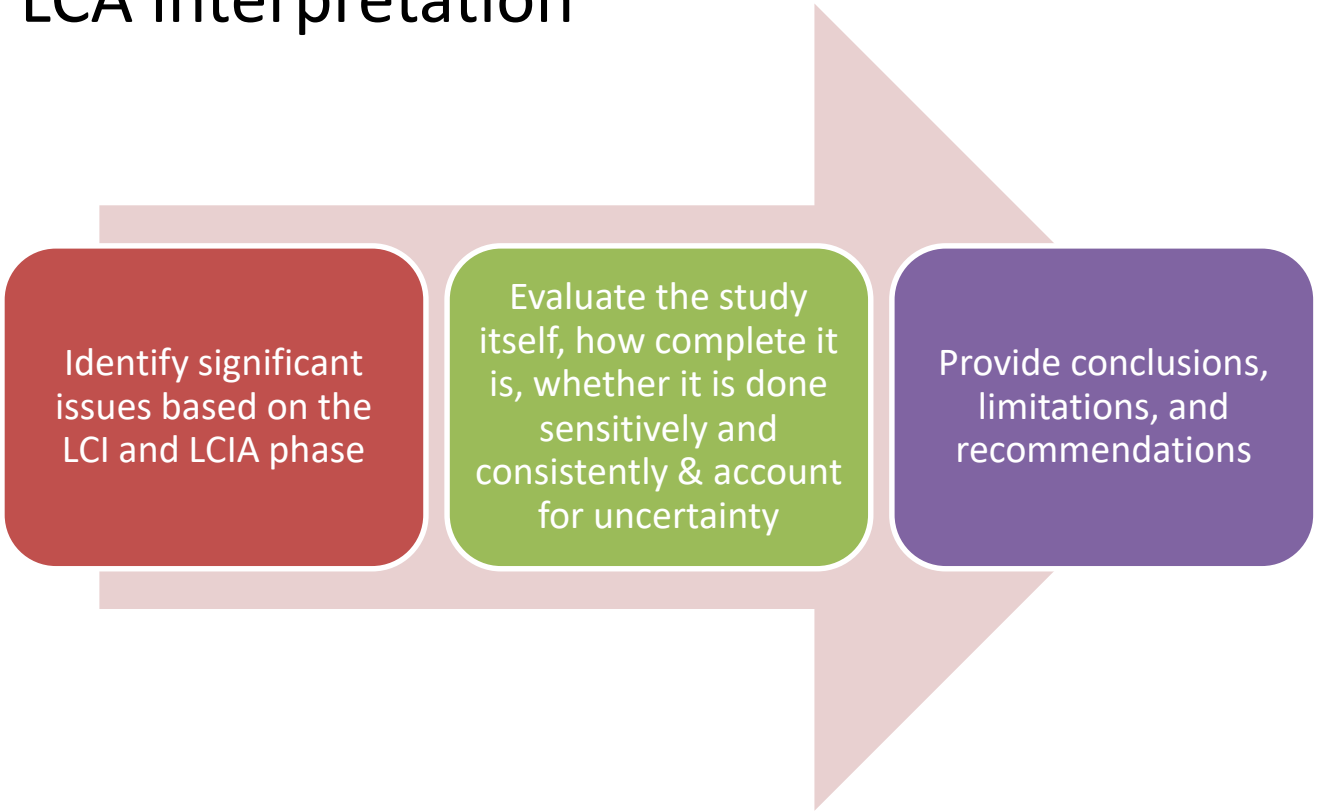
Example: Impact Assessment

Indicator	Characterization	Potential midpoint impacts	Possible endpoints
Global warming potential (GWP)	kg CO_2 equivalent	Potential for climate change based on the radiative forcing of chemicals	Sea level change, coastal area damage, agricultural effects, forest damage, plant and animal effects
Acidification potential	kg SO_2 equivalent	Potential to cause acid deposition (ex: acid rain)	Damage to forests and other plant and animal ecosystem effects, damage to buildings
Eutrophication potential	kg N equivalent	Potential to overly enrich a body of water inducing an excessive growth of plants and algae	Algae blooms, water dead zones, odors and recreational effects, human health impacts
Smog creation potential	kg O_3 equivalent	Potential to visibly pollute the air	Human health and asthma effects, damage to crops and other plants
Ozone depletion potential	kg CFC-11 equivalent	Potential to destroy the Earth's ozone layer based on a chemical's reactivity and life.	Increased UV damage causing skin cancer, cataracts, material degradation, and crop damage

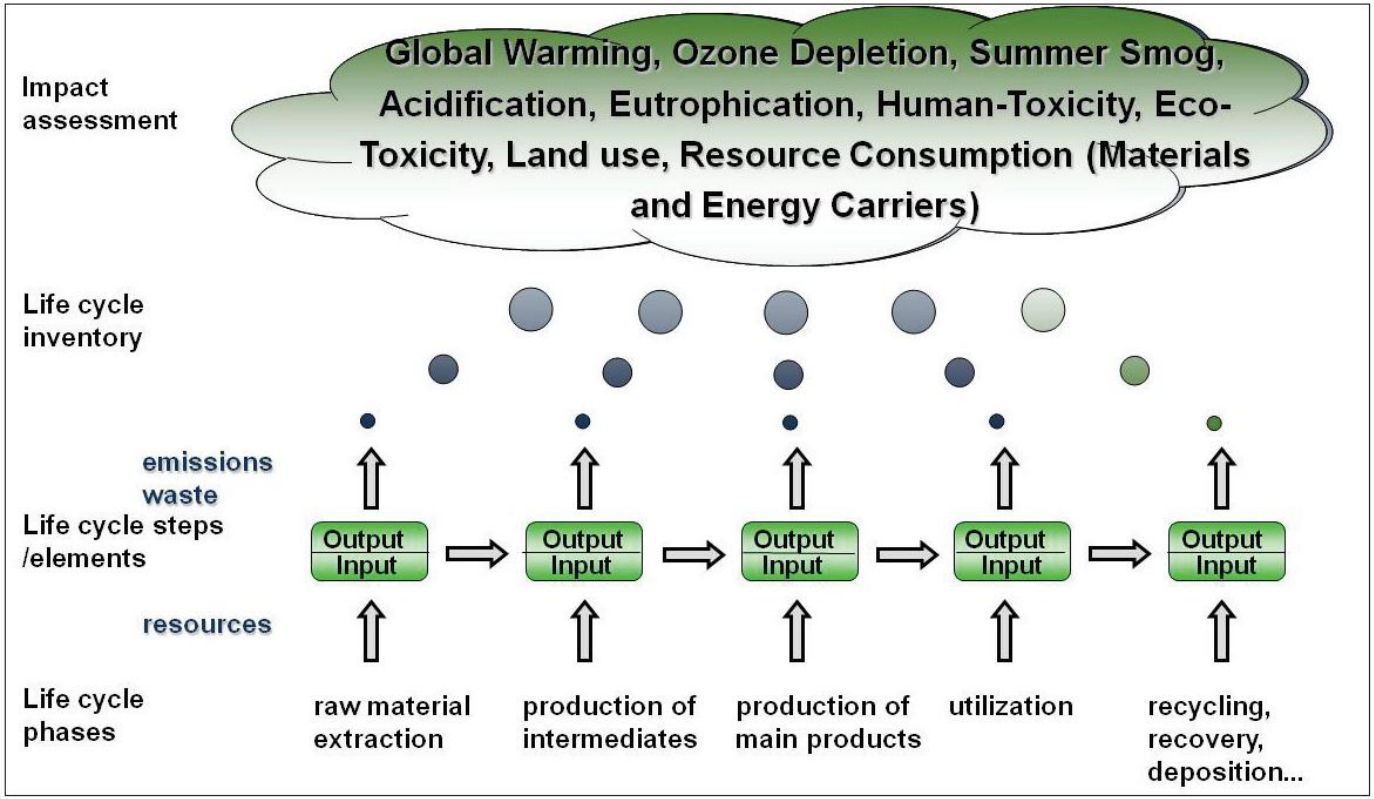
Different output of production process are linked with environmental impacts.

All outputs are converted into few categories and impacts are assessed.

LCA interpretation



Overview of LCA process



Conclusions

- Life cycle assessment is method to assess the impact of a product or service on environment.
- LCA can full fill many objective from alternative selection to marketing.
- LCA is a multi step process including goal & scope, inventory, assessment & interpretation.

References

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Thank You